

Phishing Email Investigation & Threat Intelligence

Header analysis, IOC extraction, and enrichment against a controlled phishing sample

PROJECT DOCUMENTATION REPORT

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Career track: SOC Analyst (entry-level)

Project type: Controlled phishing-email investigation

Repository: github.com/dhirajantre/phishing-email-investigation-threat-intelligence

Lab Safety: the email, domains, and IP address used in this project are synthetic test data. No real recipient, production system, or real attacker infrastructure was targeted.

Executive Summary

This report documents a controlled phishing-email investigation designed to practice an entry-level SOC Analyst workflow. A synthetic email sample was created containing common phishing indicators: brand impersonation, an urgent account-verification request, a suspicious-looking verification URL, mismatched sender/reply-to context, and failed SPF, DKIM, and DMARC authentication results.

The investigation extracted email, domain, URL, and IP indicators, enriched them against real threat-intelligence platforms, and assessed the available evidence without claiming a real-world compromise. The email is classified as a high-suspicion phishing sample for laboratory purposes and mapped to MITRE ATT&CK T1566.002 (Phishing: Spearphishing Link).

Area	Result
Email Analysis	Completed
IOC Extraction	Completed
Authentication Analysis	SPF / DKIM / DMARC failures identified
Phishing Indicators	Multiple indicators identified
Threat Intelligence	VirusTotal, urlscan.io, AbuseIPDB, and header verification checked; no real-world malicious reputation found
MITRE ATT&CK	T1566.002 - Phishing: Spearphishing Link
Compromise Evidence	None collected

1. Project Objective

Investigate a phishing email as a SOC analyst would: identify suspicious characteristics, extract indicators of compromise (IOCs), assess authentication results, perform threat-intelligence enrichment against real platforms, map the behavior to MITRE ATT&CK, and produce a documented analyst assessment.

2. Synthetic Email Sample

Sender	security@micr0soft-security.example
Reply-To	account-verification@micr0soft-security.example
Domain	micr0soft-security.example
Verification URL	https://login-micr0soft-security.example/verify
Source IP	192.0.2.10
SPF	FAIL
DKIM	FAIL
DMARC	FAIL

The **.example** domain and **192.0.2.10** address are reserved, non-production values (RFC 2606 / RFC 5737) used as synthetic documentation/test data. They are not evidence of real malicious infrastructure.

3. Email Header Analysis

The header was reviewed for sender identity, routing information, reply handling, and email authentication results.

Key Observations

- The sender domain uses **micr0soft** with a zero in place of the letter o, creating a brand-impersonation indicator.
- The Reply-To address points to the same suspicious-looking domain but a different mailbox than the visible sender.
- The Received header identifies the synthetic source address 192.0.2.10.
- SPF, DKIM, and DMARC are all shown as FAIL in the synthetic Authentication-Results header.
- The message uses urgent account-suspension language with a 24-hour deadline.

phishing_email_sample.eml

File Edit View

From: Microsoft Security <security@micr0soft-security.example>
To: employee@example.test
Subject: URGENT: Your Microsoft 365 Account Will Be Suspended
Date: Wed, 16 Sep 2026 09:15:22 +0530
Reply-To: account-verification@micr0soft-security.example
Return-Path: <account-verification@micr0soft-security.example>
Message-ID: <20260916091522.12345@micr0soft-security.example>

Received: from mail.example.test (192.0.2.10)
by mail.example.test with ESMTP;
Wed, 16 Sep 2026 09:15:20 +0530

Authentication-Results: mail.example.test;
spf=fail smtp.mailfrom=micr0soft-security.example;
dkim=fail;
dmarc=fail

MIME-Version: 1.0
Content-Type: text/html

<html>
<body>

<p>URGENT SECURITY NOTIFICATION</p>

<p>Dear User,</p>

<p>We detected unusual activity on your Microsoft 365 account.</p>

<p>Your account will be suspended unless you verify your identity immediately.</p>

<p>

VERIFY YOUR ACCOUNT

</p>

<p>If you do not complete verification within 24 hours, access to your account may be restricted.</p>

<p>Microsoft Security Team</p>

</body>
</html>

Ln 24, Col 44 | 1,160 characters | Plain text | 80% | Windows (CRLF) | UTF-8

Figure 1. Synthetic phishing email header and body evidence.

4. IOC Extraction

IOC Type	Indicator
Sender Email	security@micr0soft-security.example
Reply-To	account-verification@micr0soft-security.example
Domain	micr0soft-security.example
URL	https://login-micr0soft-security.example/verify

IOC Type	Indicator
Source IP	192.0.2.10

5. Threat Intelligence Enrichment

Each extracted indicator was checked against real threat-intelligence platforms, following the same enrichment workflow a SOC analyst would use in production triage.

5.1 Domain & URL Reputation - VirusTotal

Both the domain and the full verification URL were searched in VirusTotal. Neither returned a match, since the .example domain has never resolved on the public internet and carries no scan history. An absent VirusTotal record is the expected result for a synthetic, non-resolving domain - it confirms the indicator has no real-world footprint, rather than indicating the check was skipped.

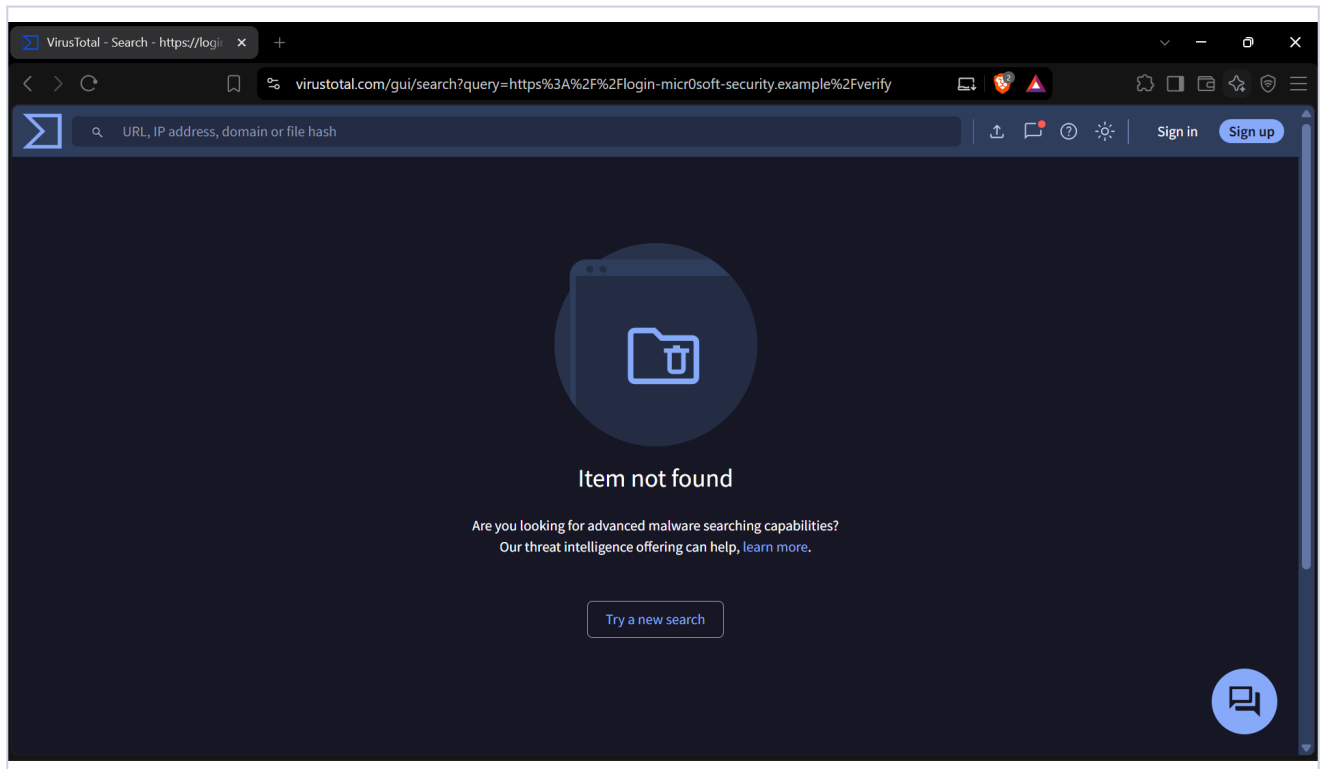


Figure 2. VirusTotal search for the verification URL - no matching item found.

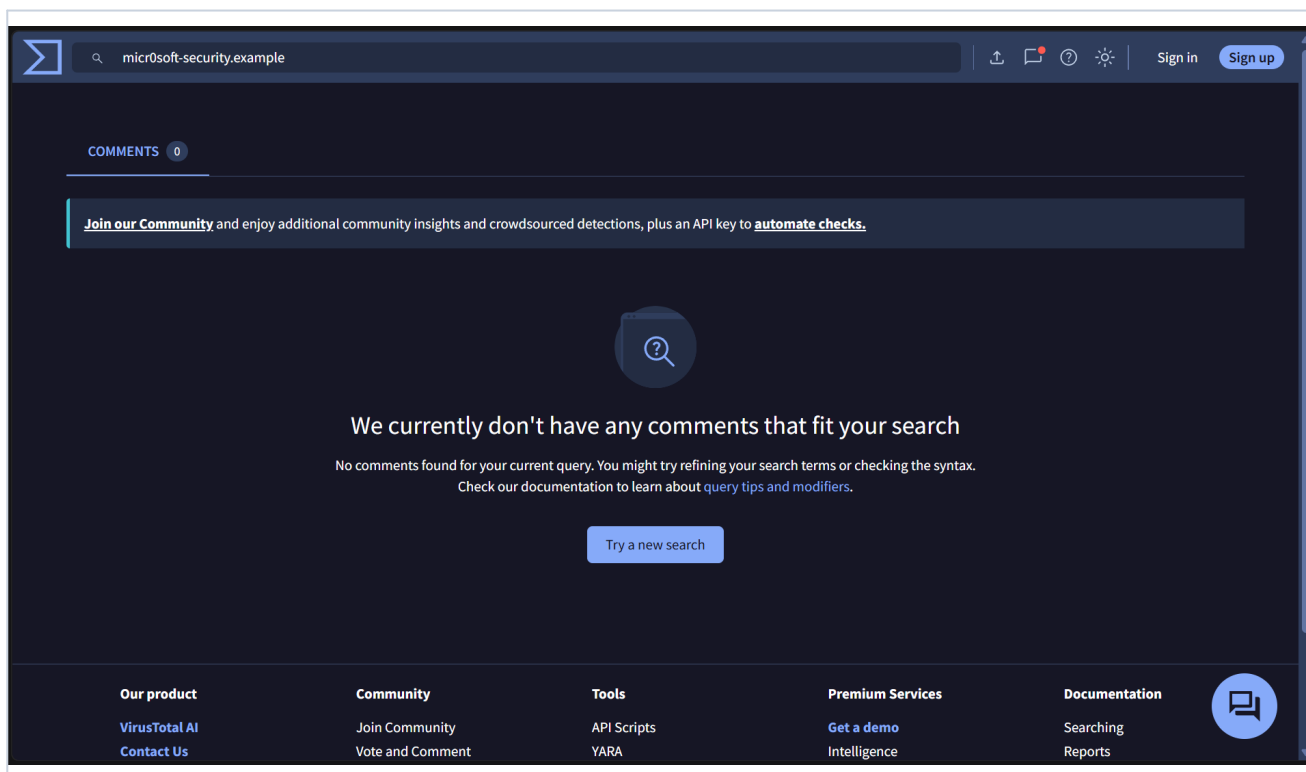


Figure 3. VirusTotal search for the sender domain - no community data on record.

5.2 URL Scan - urlscan.io

The verification URL was submitted to urlscan.io for a live scan. The scan failed with a DNS resolution error, confirming the domain does not resolve to any IPv4/IPv6 address - independent confirmation that the domain is a non-existent, synthetic value rather than live phishing infrastructure.

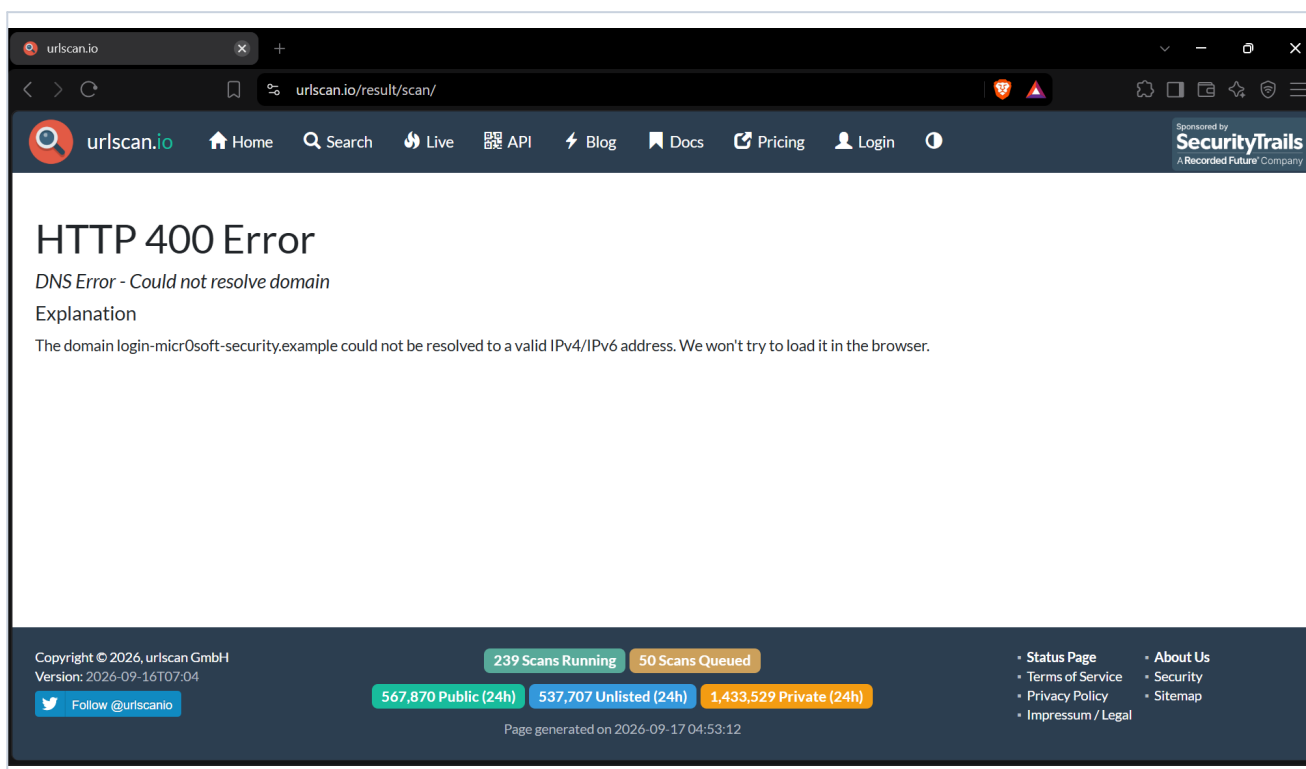


Figure 4. urlscan.io scan result - DNS error, domain could not be resolved.

5.3 IP Reputation - AbuseIPDB

The source IP 192.0.2.10 was checked against AbuseIPDB. AbuseIPDB returned 11 historical abuse reports from 5 sources, categorized as brute-force, SSH, DNS compromise, DDoS, and web-app attack activity.

AbuseIPDB's own page flags this address as reserved for documentation/testing (RFC 5737 TEST-NET-1) and states explicitly that any abuse reports against it come from internal misconfiguration or other users' testing, not real attacker infrastructure - the platform itself presents these particular reports "for entertainment and testing purposes only." This independently corroborates the lab's synthetic framing: the report count reflects other analysts and tools exercising the same test address, not a real malicious actor.

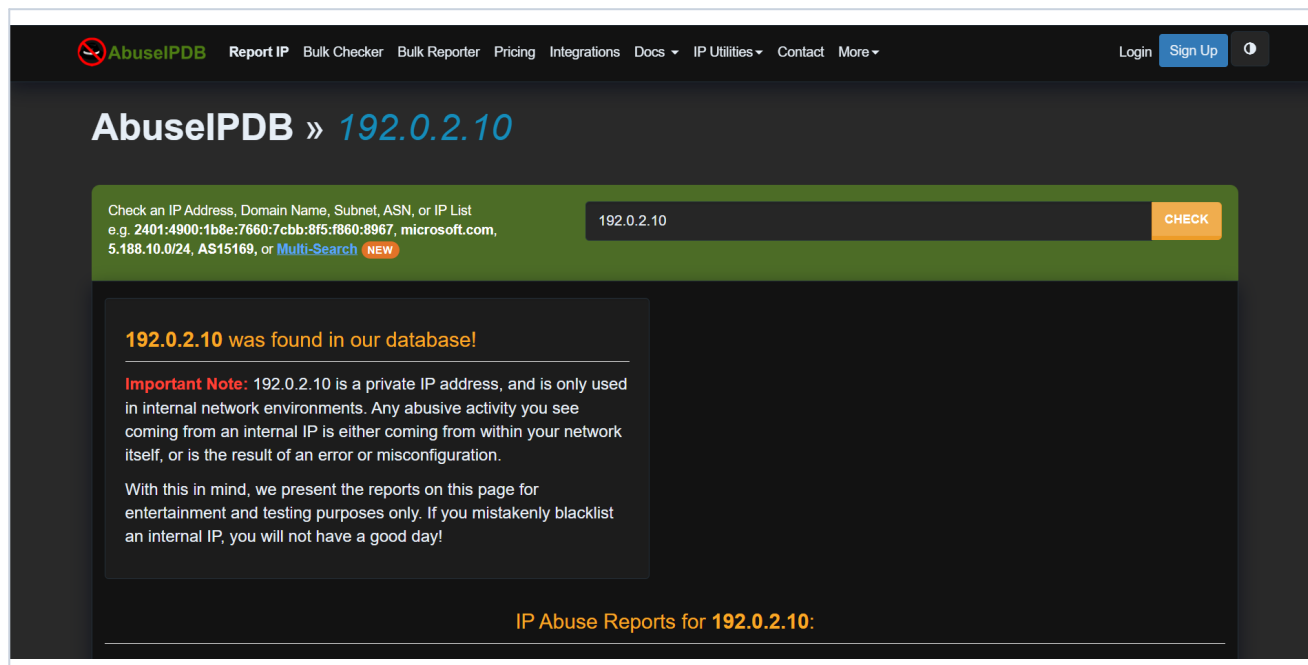


Figure 5. AbuseIPDB lookup for 192.0.2.10, including the platform's own reserved-address notice.

5.4 Header Verification - Google Admin Toolbox (Messageheader)

The corrected raw headers were run through Google's Messageheader tool as an independent, tool-based check on the manual review in Section 3. The tool confirmed all three authentication results as failed, matching the synthetic Authentication-Results header exactly:

- SPF: fail (with IP Unknown)
- DKIM: fail (with domain Unknown)
- DMARC: fail

The "Unknown" IP and domain values occur because the tool attempts to resolve mail.example.test against the live internet and cannot, since it is a reserved test hostname rather than a real mail server - a further, independent confirmation that the sample is synthetic rather than a real delivered message.

Google Admin Toolbox Messageheader

Help

MessageId20260916091522.12345@microsoft-security.example

Created at:

9/16/2026, 9:15:22 AM GMT+5:30 (Delivered after -2 sec)

From:

Microsoft Security <security@microsoft-security.example>

To:

employee@example.test

Subject:

URGENT: Your Microsoft 365 Account Will Be Suspended

SPF:

fail with IP Unknown!
[Learn more](#)

DKIM:

fail with domain Unknown!
[Learn more](#)

DMARC:

fail
[Learn more](#)

#	Delay	From *	To *	Protocol	Time received
0	-2 sec	mail.example.test	→ mail.example.test	ESMTP	9/16/2026, 9:15:20 AM GMT+5:30

ANALYZE ANOTHER HEADER

SHOW RAW HEADER

Check the [Help Center](#) for additional troubleshooting advice. If you are a Google Workspace Admin, check the documentation about [Email routing and delivery](#).

* Please note that the hostnames reported by this tool are extracted from the mail header submitted by you and no attempt has been made to verify them.

Home Feedback Google Privacy & Terms

Change language: English

Figure 6. Google Admin Toolbox Messageheader output for the corrected header block.

Threat Intelligence Summary

Indicator	Platform	Result
Verification URL	VirusTotal	No record found
Sender domain	VirusTotal	No record found
Verification URL	urlscan.io	DNS resolution failed
Source IP 192.0.2.10	AbuseIPDB	Flagged reserved/test address; 11 reports, not attributable to real attacker infrastructure
Email headers	Google Admin Toolbox	SPF/DKIM/DMARC fail confirmed; source host unverifiable (consistent with synthetic domain)

6. Phishing Risk Assessment

Indicator	Assessment
Brand impersonation	Suspicious

Indicator	Assessment
Sender domain	Suspicious
Sender / Reply-To context	Suspicious
SPF	FAIL
DKIM	FAIL
DMARC	FAIL
Urgent account-suspension language	Suspicious
Account verification request	Suspicious
Verification URL	Suspicious-looking

7. MITRE ATT&CK Mapping

Tactic	Technique	Sub-technique
Initial Access	T1566 - Phishing	T1566.002 - Spearphishing Link

The synthetic email contains a link presented as an account-verification action, supporting the mapping to T1566.002. T1204.001 (User Execution: Malicious Link) is not asserted, since there is no evidence in this lab that the link was clicked.

8. Analyst Assessment

The email contains multiple phishing indicators: brand impersonation, a suspicious verification URL, sender/reply-to concerns, and failed SPF/DKIM/DMARC results, reinforced by urgent account-suspension language. Threat-intelligence enrichment found no real-world reputation data for the domain, URL, or IP, consistent with a lab-generated sample rather than live attacker infrastructure.

FINAL CLASSIFICATION: PHISHING EMAIL -- HIGH SUSPICION (CONTROLLED LAB SAMPLE)

Compromise Assessment

No evidence was collected showing that:

- The URL was clicked
- Credentials were submitted
- Malware was executed
- An endpoint or account was compromised

No confirmed compromise identified. The investigation remains limited to the evidence available from the synthetic email sample.

9. Recommended SOC Response

If similar activity were detected in a production environment, a SOC analyst could:

- 1 Quarantine or flag the email according to the organization's email-security procedures.
- 2 Review the sender, Reply-To, domain, URL, and authentication results.
- 3 Search the environment for other emails containing the same indicators.
- 4 Identify other recipients who received the message.
- 5 Determine whether any recipient interacted with the URL.
- 6 Check whether credentials were submitted.
- 7 If interaction occurred, investigate the affected endpoint and account.
- 8 Block confirmed malicious indicators according to organizational procedures.
- 9 Monitor affected accounts and endpoints for follow-on activity.
- 10 Document and escalate the incident according to incident-response policy.

10. SOC Investigation Workflow

```
Phishing Email
|
v
Email Header Analysis
|
v
Sender / Reply-To Analysis
|
v
SPF / DKIM / DMARC Analysis
|
v
IOC Extraction
|
v
URL / Domain / IP Analysis
|
v
Threat Intelligence
|
v
Risk Assessment
|
v
MITRE ATT&CK Mapping
|
v
Analyst Assessment
|
v
SOC Response
|
v
Incident Documentation
```

11. Skills Demonstrated

Email Security

- Phishing Email Analysis
- Email Header Analysis
- Sender / Reply-To / Return-Path Analysis
- SPF, DKIM, and DMARC Analysis

Threat Intelligence

- IOC Extraction
- URL Analysis (VirusTotal, urlscan.io)
- Domain Analysis (VirusTotal)
- IP Reputation Analysis (AbuseIPDB)
- Header Verification (Google Admin Toolbox)
- Indicator Enrichment

SOC Operations

- Phishing Triage
- Security Investigation
- Risk Assessment
- Incident Documentation
- SOC Response Recommendations

Security Framework

- MITRE ATT&CK - T1566 Phishing
- T1566.002 - Spearphishing Link

Key Learning Outcomes

This project reinforced practical skills in:

- Investigating phishing email headers and authentication results
- Extracting email-based IOCs
- Applying a real threat-intelligence enrichment workflow across multiple platforms
- Interpreting an absent or reserved-address result as a meaningful finding, not a dead end
- Mapping phishing activity to MITRE ATT&CK
- Distinguishing suspicious activity from confirmed compromise
- Documenting SOC investigation findings in a structured workflow

Lab Disclaimer: This project uses synthetic phishing-email data for cybersecurity learning and SOC investigation practice. The .example domain and documentation IP address are non-production test indicators. No real person, organization, production system, or real attacker infrastructure was targeted. No real credentials were collected or submitted.

12. Project Status: Completed

[x] Synthetic Phishing Email

[x] IOC Extraction

[x] Phishing Risk Assessment

[x] Analyst Assessment

[x] Evidence Screenshots

[x] Email Header Analysis

[x] Threat Intelligence Enrichment

[x] MITRE ATT&CK Mapping

[x] SOC Response Recommendations

[x] Documentation Report

13. Conclusion

This project demonstrates a practical phishing-email investigation workflow: header and authentication analysis, IOC extraction, enrichment against real threat-intelligence platforms, MITRE ATT&CK mapping, analyst classification, and recommended SOC response. The work is intentionally framed as a controlled laboratory exercise and does not claim a real-world compromise.

github.com/dhirajantre/phishing-email-investigation-threat-intelligence